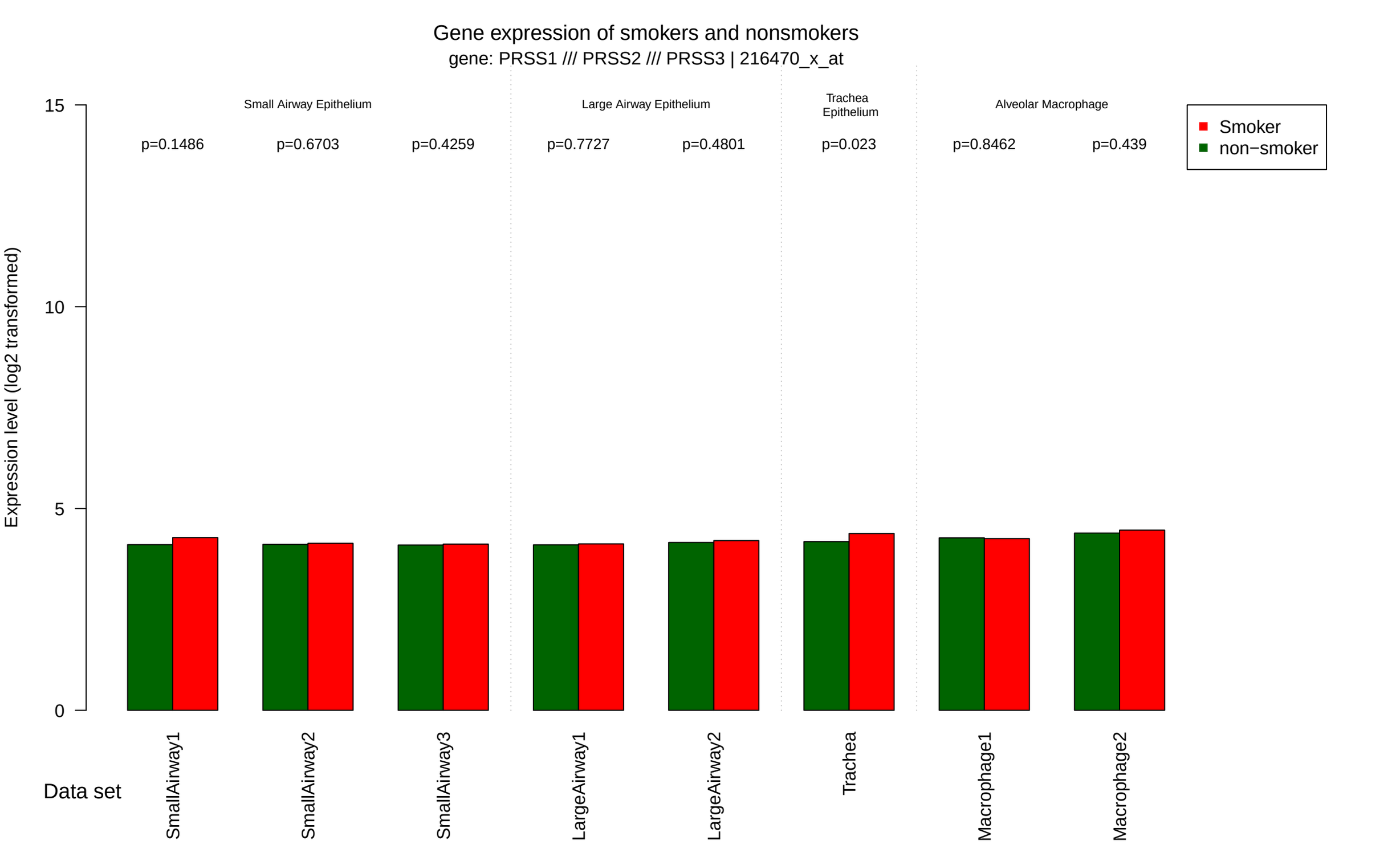
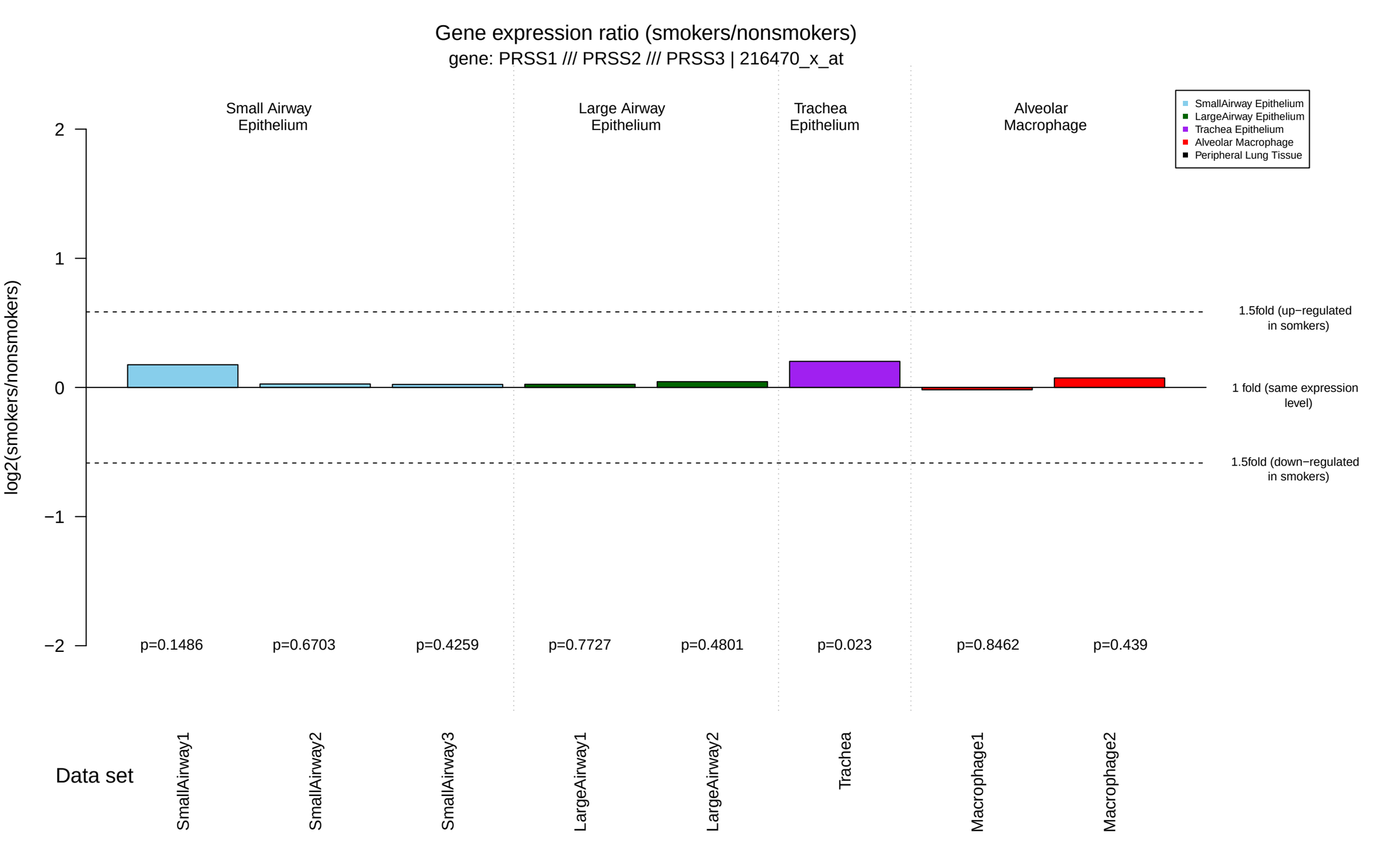
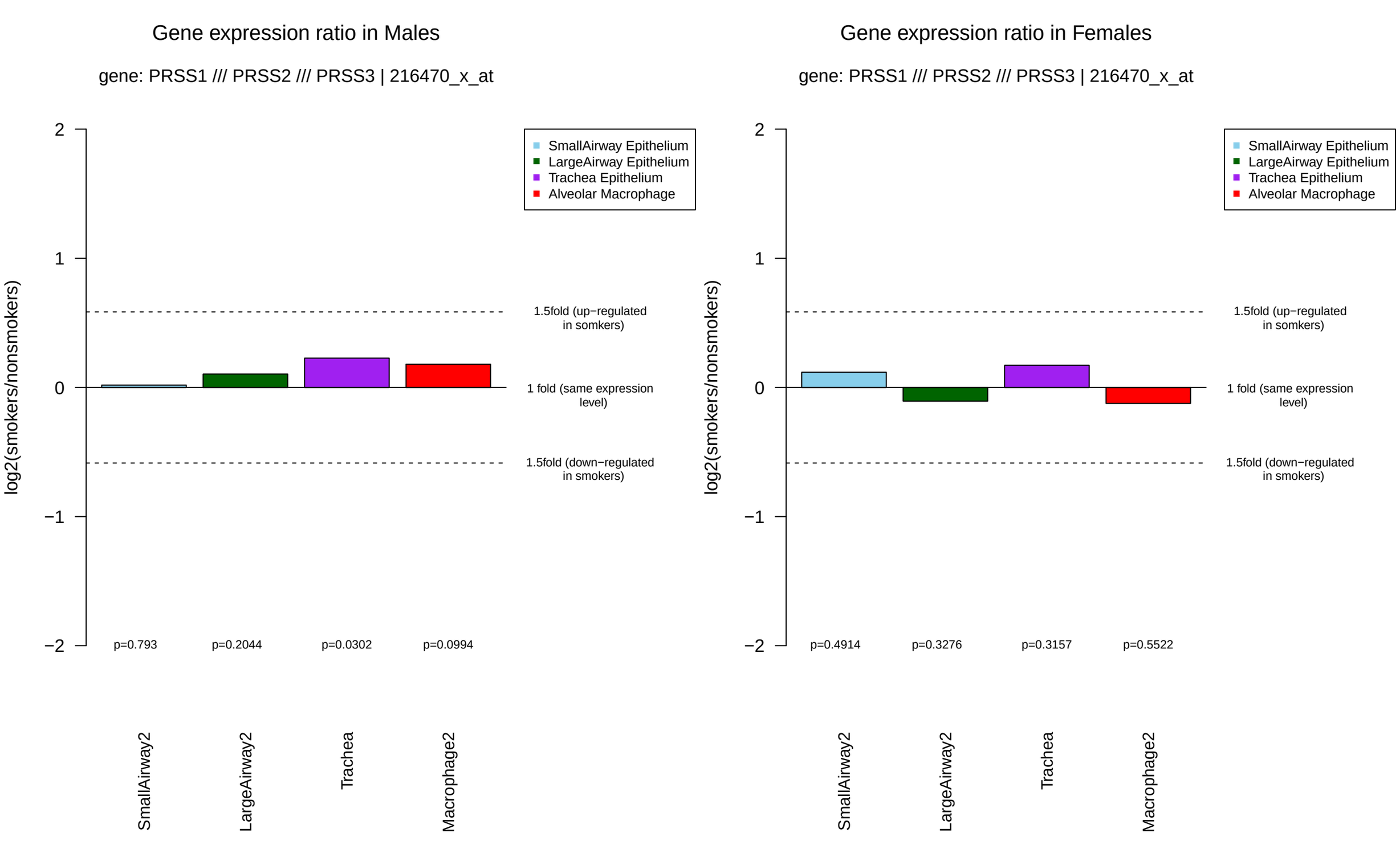
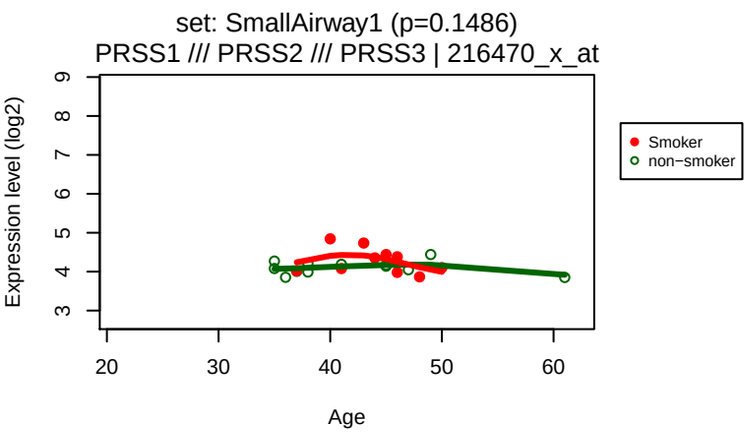
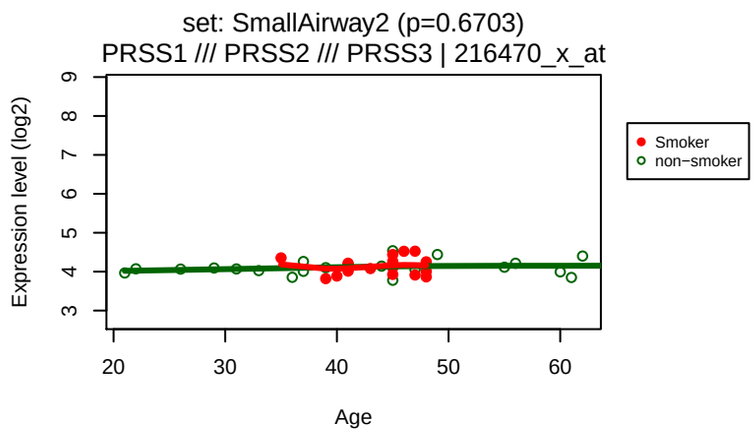
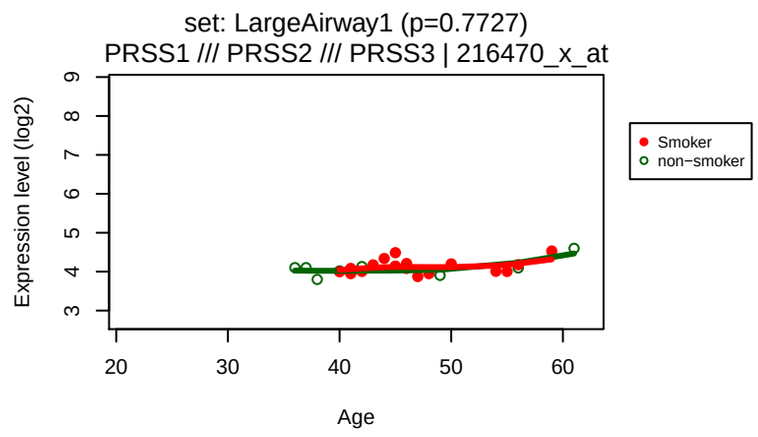

Gene expression of smokers and nonsmokers
gene: PRSS1 /// PRSS2 /// PRSS3 | 216470_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1486
p=0.6703
p=0.4259
p=0.7727
p=0.4801
p=0.023
p=0.8462
p=0.439
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

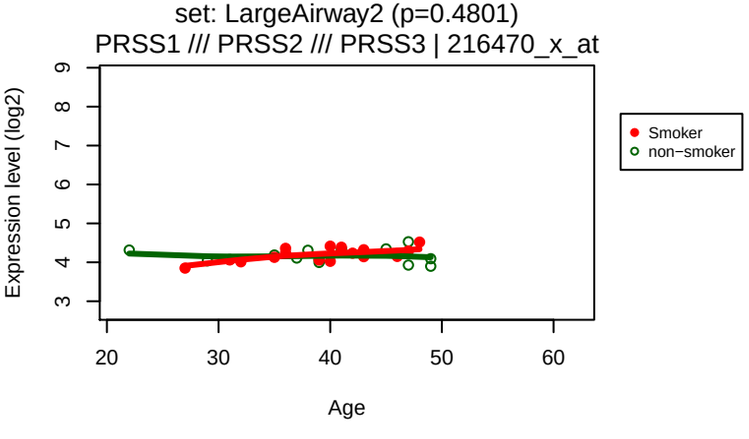

Gene expression ratio (smokers/nonsmokers)
gene: PRSS1 /// PRSS2 /// PRSS3 | 216470_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1486
p=0.6703
p=0.4259
p=0.7727
p=0.4801
p=0.023
p=0.8462
p=0.439
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

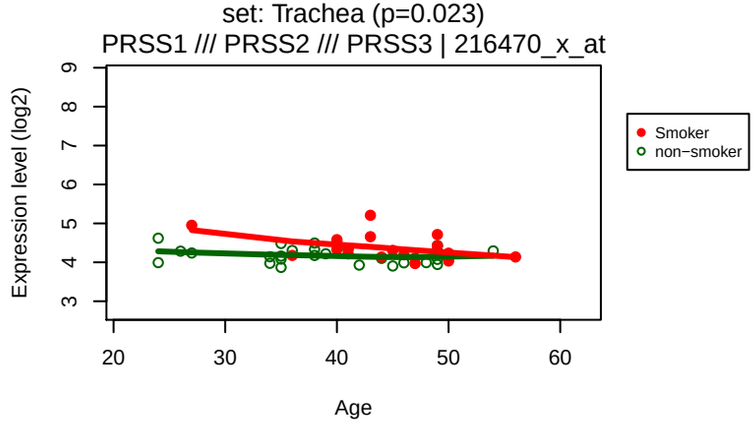

Gene expression ratio in Males
gene: PRSS1 /// PRSS2 /// PRSS3 | 216470_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.793
p=0.2044
p=0.0302
p=0.0994
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRSS1 /// PRSS2 /// PRSS3 | 216470_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4914
p=0.3276
p=0.3157
p=0.5522
SmallAirway2
LargeAirway2
Trachea
Macrophage2

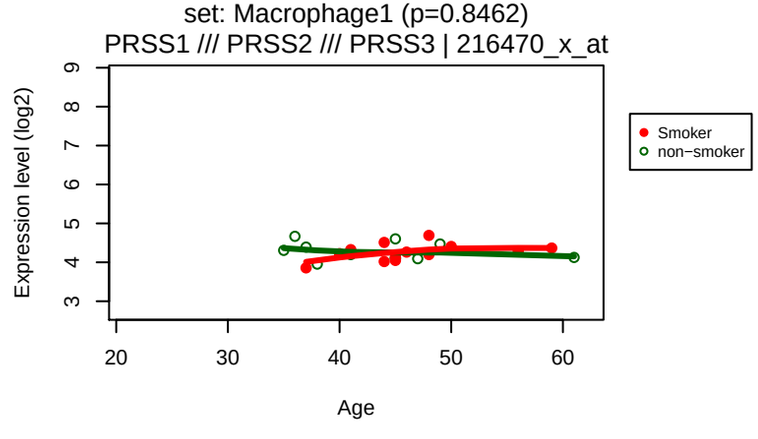

set: SmallAirway1 (p=0.1486)
PRSS1 /// PRSS2 /// PRSS3 | 216470_x_at
Expression level (log2)
Age
Smoker
non−smoker

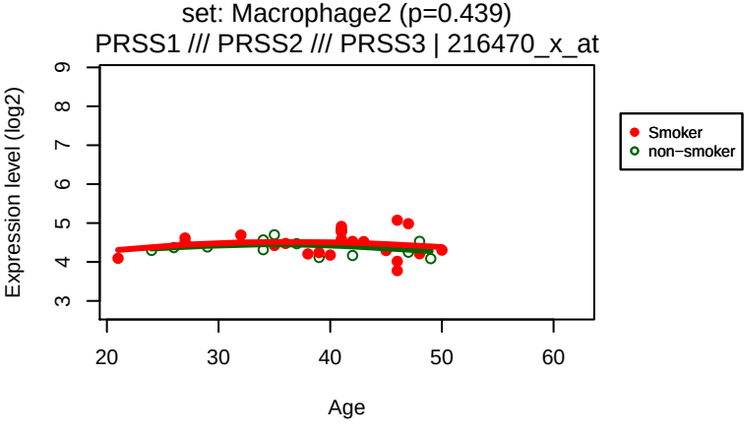

set: SmallAirway2 (p=0.6703)
PRSS1 /// PRSS2 /// PRSS3 | 216470_x_at
Expression level (log2)
Age
Smoker
non−smoker

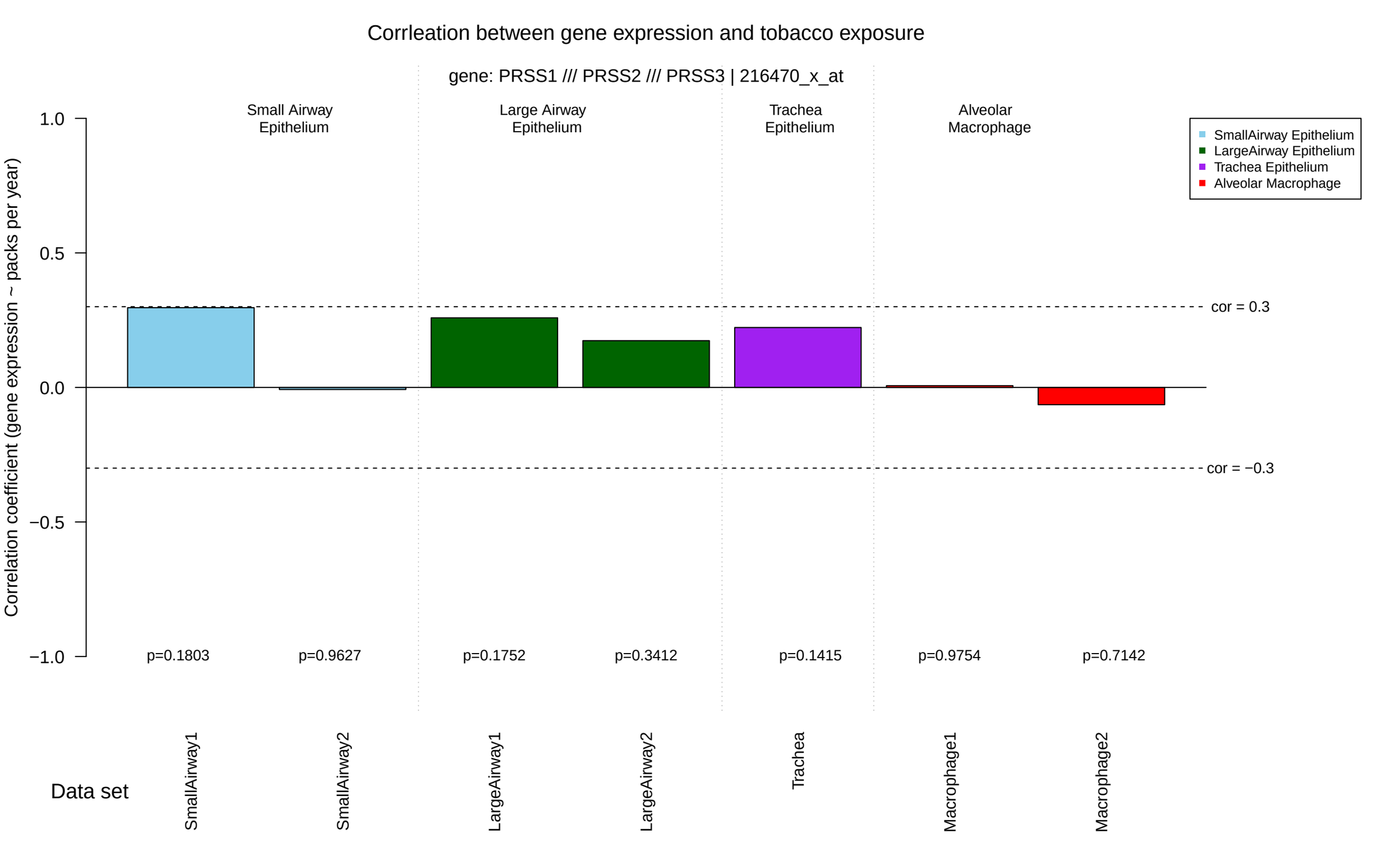

set: LargeAirway1 (p=0.7727)
PRSS1 /// PRSS2 /// PRSS3 | 216470_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4801)
PRSS1 /// PRSS2 /// PRSS3 | 216470_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.023)
PRSS1 /// PRSS2 /// PRSS3 | 216470_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8462)
PRSS1 /// PRSS2 /// PRSS3 | 216470_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.439)
PRSS1 /// PRSS2 /// PRSS3 | 216470_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRSS1 /// PRSS2 /// PRSS3 | 216470_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1803
p=0.9627
p=0.1752
p=0.3412
p=0.1415
p=0.9754
p=0.7142
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2